LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation and constructional details of rotational speed measurement
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation and constructional details of rotational speed measurement. Principle of operation and constructional details. Practical on rotational speed measurement.
5. Specific Objectives	15.1.6.1 Trainee will be able to explain construction and working of RPM indicator.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of RPM indicator calibration and reading verification; Safety: Ensure generator is de-energized before handling; Instructor supervision: Monitor trainee interaction with measurement instruments.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the function of the RPM indicator in an aircraft?
(b) Where is the RPM indicator typically located in the cockpit?
(c) Why is accurate rotational speed measurement critical for engine operation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of rotational speed measurement • Construction details of RPM indicator • Working mechanism of generator excitation • Signal transmission to cockpit display • Calibration and zeroing procedures	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of sensor types • Interpretation of tachometer signals • Troubleshooting common faults • Impact of temperature on readings • Maintenance requirements for sensors	L & Ds	PPT, PDF	20 min
(c)	• Hands-on demonstration of RPM measurement • Verification of indicator accuracy • Recording of practical observations • Comparison of theoretical vs actual values • Documentation of practical findings	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the working principle of the RPM indicator in your own words.
(b) Describe the constructional details of the RPM indicator and its components.
(c) How would you troubleshoot a faulty RPM indicator reading?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between the RPM indicator discussed in this lesson and the electrical tachometer system to be covered in LP 15.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to Electrical tachometer system	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min